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Review of Advances in Three-Dimensional Visual Perception of Spatial Objects Based on Deep Learning

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***Abstract*—Three-dimensional visual perception is a pivotal technology for achieving refined in-orbit manipulation of spatial objects and a key research direction for the intelligence of in-orbit servicing technologies. It encompasses critical techniques such as pose estimation, 3D reconstruction, and 3D generation, which are essential prerequisites for enhancing the intelligent perception capabilities of satellites. This paper aims to systematically review the research progress in 3D reconstruction and generative reconstruction technologies for spatial objects based on deep learning. Firstly, 3D scene representations are categorized into explicit, implicit, and hybrid representations, with their definitions elaborated and the advantages and disadvantages of each type analyzed comparatively, exploring the 3D representations suitable for spatial objects. Secondly, the key technologies of 3D reconstruction and 3D generation based on deep learning in recent years are reviewed, with an in-depth study and comparative analysis of the variants and extensions of various popular models, discussing their applicability in the aerospace field. Subsequently, datasets suitable for 3D reconstruction algorithm research are listed, and the characteristics of these datasets are described, summarizing the commonly used evaluation metrics for 3D reconstruction. Finally, the application scenarios of the aforementioned computer vision technologies in spatial missions are explored, their potential in the aerospace field is discussed, and future research directions worthy of further exploration are envisioned.**

Keywords—Deep Learning; 3D Reconstruction; Generative Reconstruction; Spatial Objects; In-Orbit Servicing; Situational Awareness

I. INTRODUCTION

With the increasing frequency of human space activities, the number of spatial objects such as spacecraft, debris, and asteroids continues to grow, and the geometric structures and surface textures of these objects are becoming increasingly complex. Due to the unique nature of the space environment and the complexity of space missions, achieving precise three-dimensional intelligent perception of various spatial objects is a core prerequisite for realizing in-orbit real-time servicing, asteroid sampling and exploration, autonomous rendezvous and docking, and active debris removal. This lays the foundation for information fusion and intelligent decision-making in complex scenarios. Currently, commonly used sensors for space-based observation include LiDAR and monocular cameras. Although LiDAR and other measurement devices can directly obtain depth information of targets, they struggle to capture surface textures and often have large size, weight, and power consumption. Therefore, the primary data source for three-dimensional understanding of spatial objects remains multi-view visible light images captured by monocular cameras [1].

Due to the complexity of the real space environment, visible light imaging of spatial objects often exhibits the following characteristics: The "yin-yang face" phenomenon caused by mutual occlusion in perspective projection, making it difficult to obtain complete information about components; Low brightness and low contrast due to radiation distortion; Strong reflections caused by the reflective properties of object surfaces; Overexposure or low illumination in some images due to stringent space-based optical observation conditions; Limited camera field of view, making it difficult to track targets during certain observations, resulting in unstable image sequences, and sparse perspectives limiting the number of visible light images that can be obtained; High-speed blurring caused by the dynamic characteristics of the targets themselves. These space-based imaging characteristics constrain the efficiency and accuracy of space visual perception tasks [2].

To address these issues and obtain more precise three-dimensional features of spatial objects, researchers have proposed many targeted solutions in recent years by leveraging key technologies in the field of computer vision. These include preprocessing techniques such as low-light enhancement, denoising, super-resolution reconstruction, and dynamic deblurring of two-dimensional images, as well as methods based on deep learning such as 3D reconstruction, 3D generation, and multi-modal 3D visual large models, significantly enhancing the robustness and accuracy of space perception. Generally, 3D reconstruction is a process of digital perception and reproduction, aiming to recover three-dimensional information of targets from two-dimensional images, point clouds, or other sensor observations. It helps obtain structural, dimensional, and texture information of key components of spatial objects, assisting in state estimation of targets and serving as an important prerequisite for intelligent space operations. 3D generation is a process of reasonable inference and creation, aiming to generate results without prior knowledge based on textual descriptions or two-dimensional images, often relying on large-scale data training to produce plausible random outcomes [3].

Deep learning-based methods for spatial 3D reconstruction primarily include three mainstream approaches: Multi-View

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Stereo (MVS), Neural Radiance Fields (NeRF), and 3D Gaussian Splatting (3DGS). The input data encompasses, but is not limited to, multi-view visible light images, camera parameters, and sparse point clouds. These learning models can implicitly or explicitly learn the geometric structures and texture features of spatial objects from multi-view images, or even a single image, achieving high-precision 3D reconstruction. When the input is a video sequence containing temporal information, dynamic reconstruction of spatial 3D scenes can also be achieved. However, there is a scarcity of comprehensive reviews on three-dimensional perception of spatial targets. This paper innovatively provides a detailed algorithmic analysis in this field, thoroughly elaborating on applicable methods and conducting comparative evaluations [4].

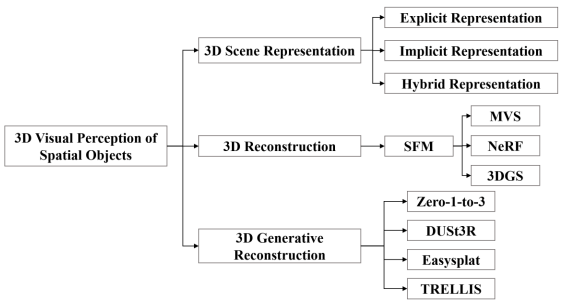

Fig. 1. Structure Diagram of the Review

In summary, deep learning-based methods have become a hot direction for solving problems related to 3D scene perception, understanding, and inference. Given the significant importance of 3D situational awareness for spatial objects, it is necessary to analyze and summarize the key technologies involved in 3D reconstruction and generation, identify current limitations, and provide recommendations for future development. However, there is a scarcity of comprehensive reviews on three-dimensional perception of spatial targets. This paper innovatively provides a detailed algorithmic analysis in this field, thoroughly elaborating on applicable methods and conducting comparative evaluations, as illustrated in Figure 1.

II. 3D Scene Representation

With the continuous iteration and updating of deep learning algorithms, 3D scene representation has also become more diverse. This section reviews key technologies such as voxel grids, point clouds, meshes, signed distance fields, neural radiance fields, 3D Gaussians, and image sequences, as illustrated in Figure 1. It summarizes the advantages, disadvantages, and application scenarios of the aforementioned 3D scene representation methods. Finally, considering the imaging characteristics of spatial objects, it explores their applicability and limitations [5].

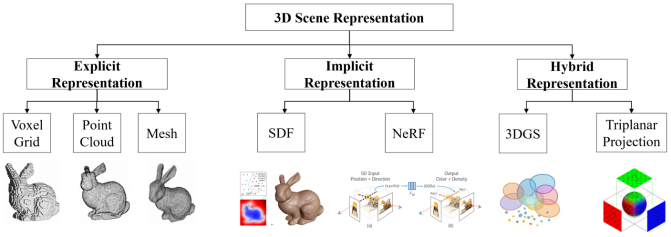

Fig. 2. Schematic Diagram of 3D Scene Representation Methods[6-8]

A. Voxel Grid

A voxel grid is a 3D array $V \in R^{H \times W \times D}$ that divides the 3D space into numerous regular, uniform grids. Each voxel stores attributes such as density, color, occupancy, or signed distance values, thereby achieving a structured volumetric scene representation. The dimensions of the array (H, W, D) represent the number of voxels in the depth, height, and width directions, respectively. The voxel data structure is well-organized, making it easy for deep learning algorithms like CNNs to extract features. The spatial occupancy information it contains is computationally efficient, but the storage and computational costs are high [6].

B. Point Cloud

A point cloud is an unordered set of N three-dimensional points $P = \{p_i \mid p_i \in R^3\}_{i=1}^{N}$, where each point is typically represented by its absolute coordinates (x, y, z) in 3D space, and sometimes augmented with additional information such as RGB color, normal vectors, or intensity. Point clouds are the most primitive form of 3D data acquisition and can be directly generated by devices like LiDAR laser scanners or depth sensors. They are characterized by properties such as sparsity, unordered nature, and storage efficiency [7].

C. Mesh

A mesh typically constructs a 3D surface using polygons $M = \{M_V, M_E, M_F\}$, where vertices M_V edges M_E, and faces M_F define the surface shape, connectivity information, and topological structure of an object. Sometimes, normals and UV texture coordinates are also included. Meshes are a compact representation of complex surface geometry, supporting texture mapping based on UV coordinates and efficient rendering. However, their data structure is irregular, and they are difficult to directly acquire from spatial environments [8].

D. Implict Neural Fields

Signed Distance Field (SDF) and NeRF are implicitly continuous functions parameterized by neural networks. Both fields and functions essentially represent a mapping relationship. Specifically, SDF maps the position $x \in R^3$ of a point in space to a signed distance $s(x) \in R$, defining the zero-level set as its surface, and is primarily rendered using sphere tracing. On the other hand, NeRF maps the position $x \in R^3$ of a point in space and the viewing direction $r \in R^3$ to volume density $\sigma(x, r) \in R^3$ and color $c(x, r) \in R^3$, and is primarily rendered using differentiable volume rendering. Unlike discrete representations such as voxels or points, implicit representations can continuously characterize a scene and synthesize high-fidelity novel view images. However, neural network inference is time-consuming, and the scene cannot be easily edited [9].

E. 3D gaussians

3D Gaussian representation explicitly models a 3D scene using N learnable Gaussian ellipsoid primitives $G = \{(\mu_i, \sum_i, c_i, \alpha_i)\}_{i=1}^{N}$. Each Gaussian ellipsoid primitive is

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defined by its center $\mu_i \in R^3$, a covariance matrix $\sum_i \in R^{3\times3}$ that includes rotation and scaling transformations, RGB color $c_i \in R^3$ and transparency $\alpha_i \in [0,1]$. The explicit representation of Gaussian ellipsoids makes the scene editable and enables efficient rendering. However, this approach tends to produce numerous tiny transparent Gaussian spheres, which can introduce "artifacts" that interfere with the reconstruction of the scene [10].

F. Triplanar Projection

Tri-plane mapping is an explicit and structured 3D scene representation method that compresses 3D spatial information and stores it in three orthogonal high-resolution 2D feature maps, corresponding to the XY, XZ, and YZ planes, respectively. These 2D feature maps function as high-dimensional feature vectors, encoding appearance, geometry, and other information for each spatial location. They are often used as intermediate representations or encoders for methods like NeRF. The hybrid representation of tri-plane mapping is highly storage-efficient, maintaining 3D consistency with implicit representations while also being compatible with 3D architectures. This approach offers strong flexibility and scalability [11].

G. Image Sequence

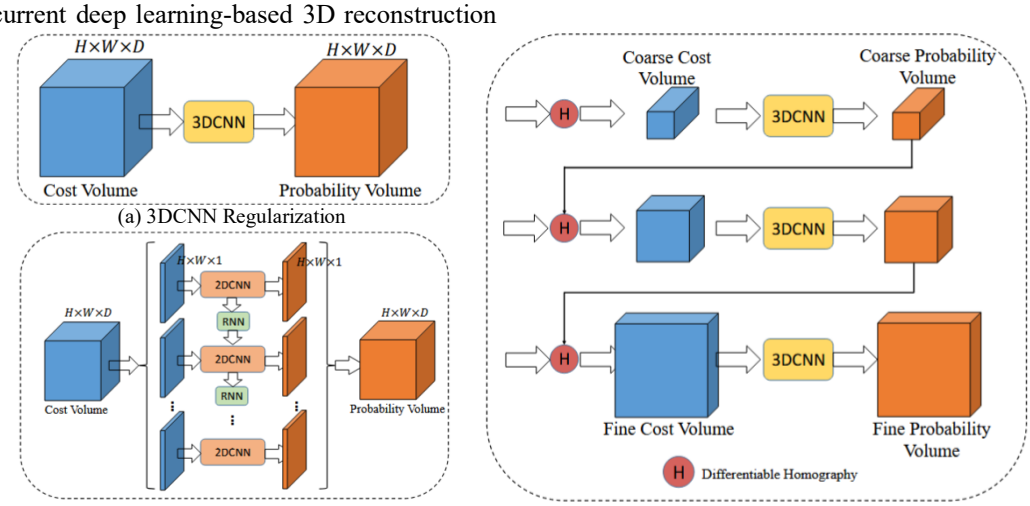
Image sequences are the most widely used representation of 3D scenes in image and video generation methods. By utilizing multiple images from different viewpoints along with their corresponding camera parameters, they implicitly encode the geometric structure of a 3D scene. For example, multi-view reconstruction methods (such as $C = \{I_i \in R^{H\times W\times 3}\}_{i=1}^{N}$) can infer 3D structures, surface textures, and other features from these image sequences. Multi-view image sequences are easy to acquire and require minimal equipment, making them a fundamental data source for generating point clouds, meshes, NeRF, 3DGS, and other 3D scene representations mentioned above.

III. Deep Learning-Based 3D Reconstruction Methods

This section summarizes the three most commonly used frameworks in current deep learning-based 3D reconstruction techniques: Structure from Motion (SfM), NeRF, and 3DGS. Throughout their development, these three frameworks—MVS, NeRF, and 3DGS—have each spawned a rich variety of methodological variants. MVS has gradually evolved from traditional PatchMatch-based algorithms to end-to-end deep learning inference models, continuously improving robustness and reconstruction completeness in challenging scenarios such as complex occlusions and weak textures. The research focus of NeRF has centered on enhancing training and rendering speeds (e.g., Instant-NGP), adapting to unbounded scenes (e.g., Mip-NeRF 360), and modeling dynamic scenes. Meanwhile, 3DGS has rapidly advanced along paths such as improving optimization stability and geometric accuracy, reducing memory consumption, and enhancing compatibility with diverse input data (e.g., SfM point clouds, monocular depth). These collective advancements have driven 3D reconstruction technology toward higher precision, greater efficiency, and stronger generalization capabilities [12]. Below, we provide a literature review of each of these frameworks.

A. MVS

Deep learning-based MVS primarily relies on post-processing of the estimated depth maps. A typical MVS model consists of three main components: the feature extraction network, the cost volume construction function, and the cost volume regularization network. In 2018, Yao and his team from the Hong Kong University of Science and Technology proposed MVSNet [13], marking a significant milestone in using deep learning to address MVS problems. MVSNet employs differentiable homography transformations to construct cost volumes for multi-view depth estimation. In 2020, CVP-MVSNet [14] introduced a pyramid-style cost volume structure, enabling layer-by-layer refinement of depth maps and achieving coarse-to-fine depth adjustment. In 2022, Wang et al. addressed issues caused by the structural symmetry and non-Lambertian imaging characteristics of spatial objects, which lead to incorrect or insufficient feature point matching, by proposing a method based on the MVSNet deep learning network. This approach leverages deep learning techniques to extract high-level semantic information from images, thereby enhancing the accuracy of 3D reconstruction for spatial targets [15].



(a) 3DCNN Regularization

(b) RNN Regularization

(c) "Coarse-to-Fine" Regularization

Fig. 3. The Structure of MVSNet[13]

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B. NeRF

NeRF demonstrates significant advantages in the task of 3D reconstruction of spatial targets. In the field of 3D reconstruction for spatial objects, traditional methods often face challenges such as strong data dependency, high computational complexity, and difficulties in handling complex lighting and occlusions. For tasks involving the 3D reconstruction of spacecraft, satellites, planetary surfaces, and similar targets, it is typically necessary to process large-scale image data, high dynamic range lighting, and occlusions. Traditional methods struggle to effectively address these issues, creating an urgent need for a novel 3D reconstruction approach that can adapt to complex scenes and exhibit high robustness. NeRF combines neural networks with a radiative transfer model, enabling it to learn depth and density information of scenes from optical images of spacecraft and achieve high-fidelity reconstruction of spacecraft surfaces [16].

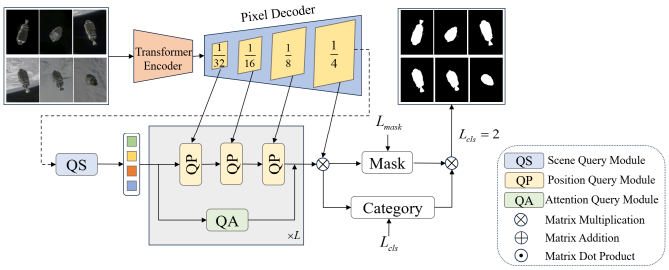

Fig. 4. Overview of the 3D satellite reconstruction pipeline in NeRF

Mip-NeRF [17] introduced a multi-scale representation method to reduce aliasing artifacts in rendered images. However, Mip-NeRF struggles to accurately simulate light reflections on object surfaces. To address this issue, Verbin proposed Ref-NeRF [18], which explicitly models view-dependent reflections and refractions in scenes. Ref-NeRF replaces NeRF's parameterization of view-dependent outgoing radiance with a representation of reflected radiance and constructs this function using a collection of spatially varying scene properties. Subsequently, Zip-NeRF [19] tackled aliasing issues in grid-based NeRF methods by employing multi-scale sampling and feature down-weighting strategies.Training neural radiance fields without image pose information is highly challenging. Some studies have shown that NeRF and camera poses can be jointly optimized in forward-facing scenes to address this problem. However, significant difficulties remain when the camera undergoes drastic motion. NoPe-NeRF [20] introduced a distortion-free monocular depth prior to solve this issue. These priors are generated by correcting scale and translation parameters during training, which constrains the relative poses between consecutive frames through a novel loss function. Additionally, pixelNeRF's multi-view projection and mean processing are straightforward and efficient, while MVSNeRF adopts a more complex multi-view stereo (MVS) approach. Despite their differences in implementation, both methods share similar principles and focus on generalization capabilities [21].

C. 3DGS

3DGS employs an explicit representation and a highly parallel workflow, which facilitates more efficient computation and rendering. Its innovation lies in the unique integration of the advantages of differentiable pipelines and point-based rendering techniques. By representing scenes with learnable 3D Gaussian functions, it retains the desirable properties of continuous volumetric radiance fields, which are crucial for high-quality image synthesis, while simultaneously avoiding the computational overhead associated with rendering empty space.

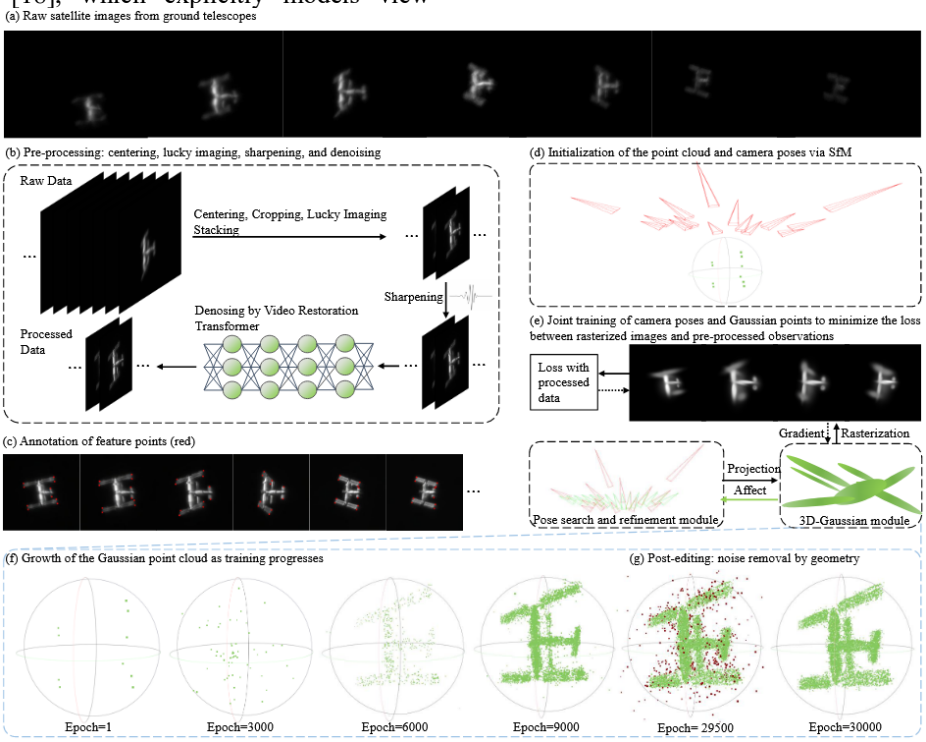

Fig. 5. Overview of the 3D satellite reconstruction pipeline in 3DGS[25]

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Tang et al. [22] accelerated the convergence process by employing progressively densified 3D Gaussians, significantly reducing computation and optimization time during generation through a more efficient sparse point cloud representation. Additionally, transforming 3D Gaussian point clouds into detailed texture UV mappings not only enhanced texture details but also streamlined the post-processing workflow, facilitating further applications. Antoine et al. [23] introduced a regularization term during the optimization of 3D Gaussian point clouds, enabling Gaussian points to align with scene surfaces, thereby improving the accuracy of mesh extraction. To address high-frequency artifacts caused by varying sampling rates (e.g., changes in focal length or camera distance), Yu et al. [24] effectively constrained the size of 3D Gaussian primitives by introducing a 3D smoothing filter, successfully eliminating high-frequency artifacts during scaling. Chang et al. [25] proposed an innovative 3D reconstruction framework that utilizes video data captured by amateur telescopes to reconstruct 3D models of low Earth orbit satellites. By integrating preprocessing, a customized structure-from-motion approach, and an improved 3D Gaussian rendering algorithm, the framework effectively tackled challenges posed by atmospheric turbulence, light pollution, and long focal lengths. Experimental results demonstrated its superior performance on both simulated and real observation data, offering a cost-effective new approach for ground-based satellite monitoring.

In summary, traditional MVS methods exhibit strong robustness but incur significant computational overhead, requiring iterative optimization through bundle adjustment and continuous projection. NeRF methods have undergone continuous optimization in rendering speed, reconstruction accuracy, and domain expansion, establishing a foundational application in the space domain, yet they still require enhancement in lightweight deployment capabilities. 3Dgs demonstrates substantial potential, promising real-time onboard rendering, and is poised to be a prominent direction for future development.

IV. Generative Novel View Synthesis

Due to the constraints of unique spatial environments and observation conditions, optical imaging of spatial targets using a monocular camera often makes it difficult to obtain a large number of images. However, existing 3D reconstruction methods heavily rely on precise camera poses and multi-view image sequences, posing significant challenges for novel view synthesis of spatial targets. As a result, many generative novel view synthesis methods under sparse viewpoints have emerged recently.

The core idea of Zero-1-to-3 is to fine-tune a pre-trained image diffusion model, aiming to provide a controllable viewpoint generation model. This method incorporates camera viewpoint parameters as conditional inputs into the pre-trained diffusion model and fine-tunes it using a large number of "input image - camera pose - new viewpoint image" triplets, with the output being a new viewpoint 2D image [26]. Dut3r utilizes a shared ViT encoder across multiple views and two Transformer decoders, one for the reference image and the other for the target image, predicting a point cloud map of size $H \times W \times 3$ for the reference image and the point cloud of the target image in the reference image's coordinate system. However, DUSt3R can only process image pairs, and multi-image processing requires post-hoc alignment of point clouds, which is time-consuming and memory-intensive [27]. EASYSPLAT proposes using grid-based uniform sampling or starting directly from random initialization, eliminating the model's dependency on SfM while optimizing Gaussian distribution densification and pruning strategies [28]. TRELLIS introduces a miniature, per-frame updated training process. Its breakthrough lies not in improving static reconstruction quality but in shifting the training process from "offline" to "online," enabling users to interact with a "learning" 3D scene in real-time, akin to playing a 3D game [29]. This has revolutionary implications for generative 3D reconstruction of spatial target optical image datasets under sparse viewpoints.

V. Conclusion

In the context of three-dimensional situational awareness of spatial targets, 3D reconstruction and 3D generation have distinct focuses in principle but exhibit complementary roles in technology. 3D reconstruction is primarily used to accurately restore the shape and structure of target objects, while 3D generation is employed to create virtual models and enhance data. In the future, with the integration and advancement of these technologies, both will find broader applications in areas such as space exploration, navigation, and science popularization.

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